HANDWRITTEN STUDY NOTES

GEOSYNTHETICS AND REINFORCED EARTH STRUCTURES

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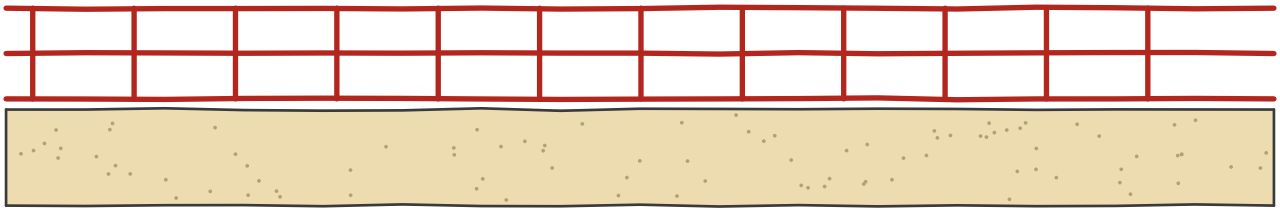
UNIT – III

Reinforced Earth Retaining Walls

Syllabus topics covered in these notes:

- Reinforced earth retaining walls: introduction, stability mechanisms
- Design of RE wall: selection of materials, geotechnical analysis
- Reinforcement layout and spacing; stability analysis
- Advantages over conventional retaining walls

soil + reinforcement = strong composite



1. Reinforced Earth Retaining Walls

A **reinforced earth (RE) / mechanically stabilised earth (MSE) wall** is a retaining structure in which a mass of compacted granular fill is reinforced by horizontal layers of strips, grids or sheets that are connected to a **facing**. The reinforced mass acts like a **gravity wall** that resists the thrust of the retained soil behind it.

Parts of an RE wall

- **Reinforced zone** – compacted backfill + reinforcement (length L, height H).
- **Retained fill / natural soil** behind the reinforced zone.
- **Facing** (precast panels, blocks, wrap-around, welded wire) and **leveling pad**.
- **Drainage** – behind and below the wall (weep holes, geocomposite, sub-surface drain).
- **Foundation soil** – must have adequate bearing capacity and limited settlement.

Types

Type	Reinforcement	Facing
Reinforced Earth (Vidal)	Galvanised steel strips	Cruciform RCC panels
Welded wire / bar-mat wall	Steel grids	Wire basket, panels
Geosynthetic wall	Geogrid / geotextile / geostrip	Blocks, wrap-around, panels
Soil nailing / anchored (different)	Passive nails	Shotcrete

2. Stability Mechanisms

Two classical ways of looking at an RE wall:

- **Tie-back wedge method** – the active wedge (Rankine/Coulomb) behind the facing tries to slide; reinforcement beyond the wedge acts as a **tie-back** (anchor). Used for **extensible** reinforcement.

- **Coherent gravity method** – the reinforced block behaves as a rigid, coherent mass; the line of maximum tension is **not** the Rankine line but a bilinear curve. Used for **inextensible** (steel) reinforcement.

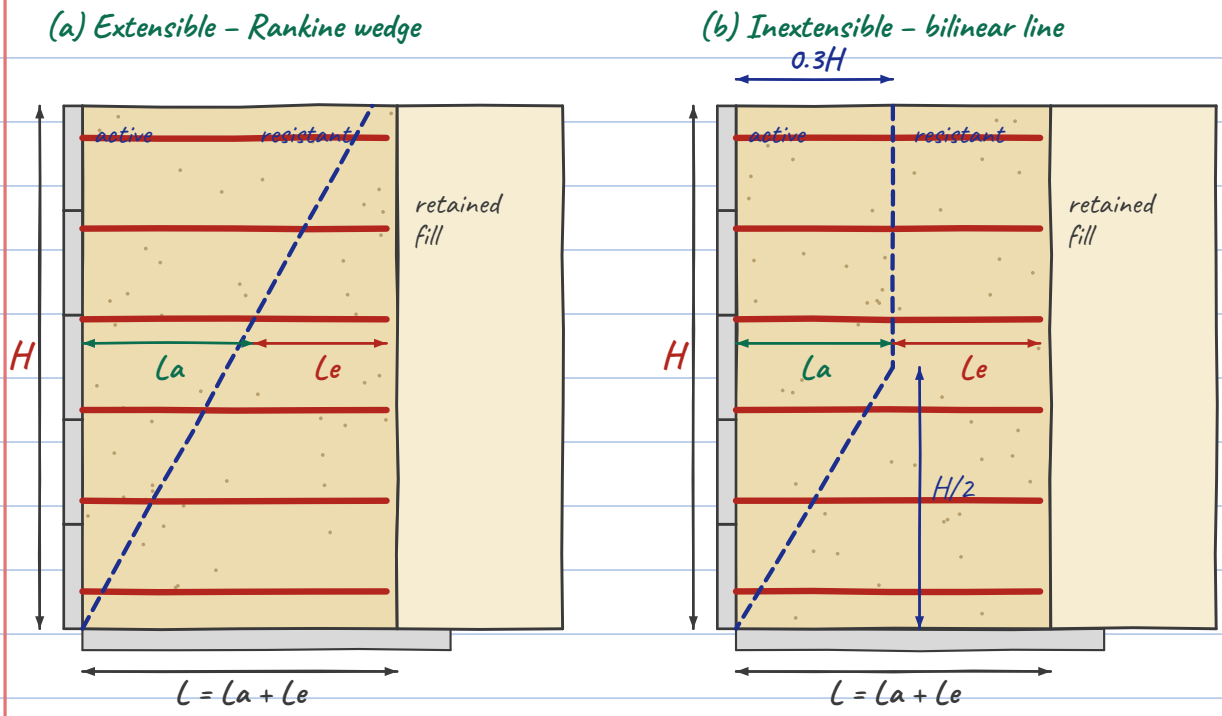


Fig 3.1 Failure surfaces and reinforcement lengths

(a) Rankine plane: $L_a = (H - z)\tan(45° - \varphi/2)$

(b) Bilinear: vertical 0.3H from face down to H/2, then straight to the toe

Mechanism

Active zone (La) → soil tries to move out and pulls the strip ⇒ tension T.

Resistant zone (Le) → soil holds the strip by friction ⇒ pull-out resistance.

Total length L = La + Le at every level.

Lateral earth pressure coefficient in the reinforced zone

Reinforcement	K at top (z = 0)	K at depth ≥ 6 m
Steel strips (original Terre Armée)	K0 = 1 – sin φ	Ka
Steel strips (AASHTO)	1.7 Ka	1.2 Ka
Steel grids / bar mats	≈ 2.5 Ka	1.2 Ka
Geosynthetics (extensible)	Ka	Ka

Reason: inextensible reinforcement prevents lateral strain so the soil stays close to the **at-rest** state near the top, where compaction-induced stresses are large; extensible sheets allow the soil to reach the active state.

Tie-back wedge vs coherent gravity

Feature	Tie-back wedge	Coherent gravity
Reinforcement	Extensible (geosynthetic)	Inextensible (steel)
Failure surface	Rankine plane $45° + \varphi/2$	Bilinear max-tension line
Pressure coefficient	Ka	K0 → Ka (1.7→1.2 Ka)
Idea	Strips act as anchors	Block acts as gravity wall
Used in	FHWA geosynthetic design	AASHTO / Reinforced Earth

3. Design of RE Wall – Steps

1. **Geometry** – wall height H, surcharge q, slopes, ground water, design life (75–100 years).
2. **Soil parameters** – reinforced fill (φ_r, γ_r), retained fill (φ_b, γ_b), foundation soil (c, φ, q_a).
3. **Select materials** (below).
4. **Trial** reinforcement length $L \geq 0.7H$ (min 2.5 m), spacing S_v, type.
5. **External stability** – sliding, overturning, bearing, global/ settlement.
6. **Internal stability** – rupture and pull-out at each layer; connection check.
7. **Facing / drainage details**, then revise and finalise.

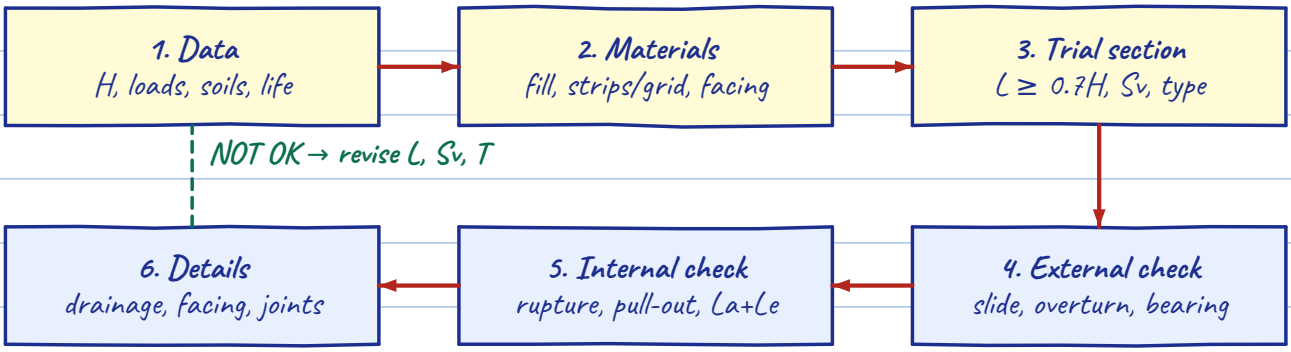


Fig 3.3 Design flow of an RE wall

3.1 Selection of materials

- **Fill:** granular, free-draining (fines ≤ 15 %, PI ≤ 6, max size ≤ 75 mm), $\varphi \geq 30°–34°$, compact to ≥ 95 % MDD. Electrochemical limits if steel is used.
- **Reinforcement:** choose by life, environment and cost: galvanised steel (strength, no creep) vs geogrid / geostrip (corrosion free). Use long-term strength $T_{al} = T_{ult}/(RF_{ID}\,RF_{CR}\,RF_{D})$.
- **Facing:** RCC panels (M30+), modular blocks, gabions, wrap-around, welded wire with vegetation – choose for height, aesthetics, tolerance to settlement.
- **Drainage:** granular blanket or geocomposite behind the reinforced zone; weep holes; toe drain; avoid water behind the wall.

3.2 Geotechnical analysis – external stability

The reinforced block (width L, height H) is treated as a **rigid gravity wall** acted on by the thrust of the retained soil.

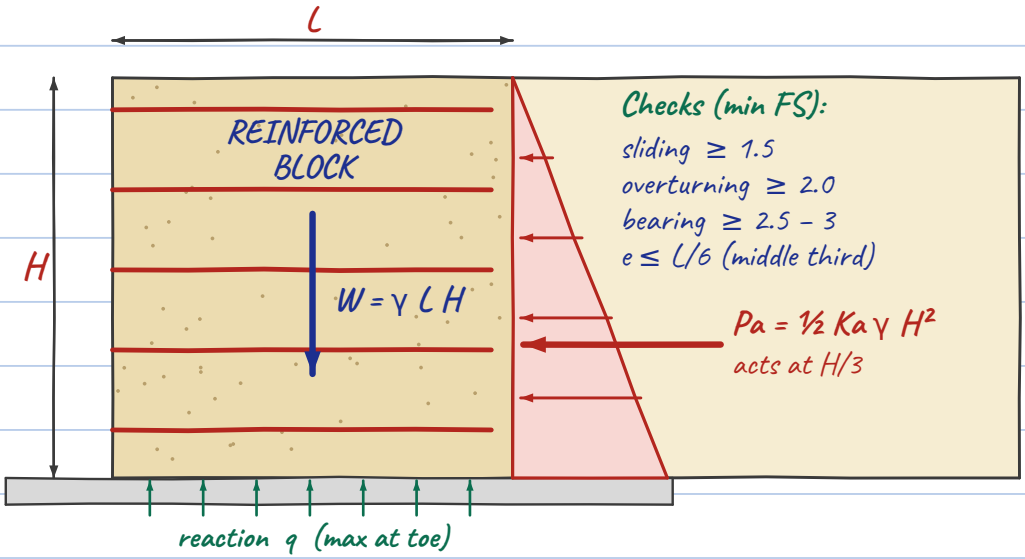


Fig 3.2 External stability of the reinforced block

$$P_a = \tfrac{1}{2} K_a \gamma_b H^2 \ (+ K_a q H \text{ for surcharge } q)$$

$$W = \gamma_r L H$$

- **Sliding:** $FS_s = W \tan \varphi_{min} / P_a \geq 1.5$ (φ_{min} = least of φ_r, $\varphi_{foundation}$, interface).
- **Overturning** about toe: $FS_o = M_R / M_O = (W \cdot L/2) / (P_a \cdot H/3) \geq 2.0$.
- **Eccentricity** $e = L/2 - (M_R - M_O)/W \leq L/6$.
- **Bearing:** $q_{max} = W / (L - 2e) \leq q_{all}$ ($FS \geq 2.5$–3 on $q_{ult} = cN_c + \tfrac{1}{2}\gamma (L - 2e) N_\gamma + \gamma D N_q$).
- **Global / deep-seated stability:** Bishop or Janbu method with reinforced block as part of slope, $FS \geq 1.3$–1.5. **Settlement:** total and differential (consolidation of foundation).

3.3 Internal stability

At depth z, the **horizontal stress** and the **tension** to be resisted by each reinforcement layer (vertical spacing S_v):

$$\sigma_h = K (\gamma z + q)$$

$$T_{max} = \sigma_h S_v \text{ (per metre run of wall for sheets / grids)}$$

$$T_{max} = \sigma_h S_v S_h \text{ (per strip, horizontal spacing } S_h)$$

- **Rupture check:** $T_{al} \geq T_{max}$ (using $FS \geq 1.5$ on allowable strength).

- **Pull-out check**: required embedment $L_e \geq FS \cdot T_{max} / (2\, b\, F^* \sigma_v\, C)$, $FS \geq 1.5$, min $L_e = 1$ m.
- Total length at each level $L = L_a + L_e$; adopt the greatest required, $L \geq 0.7H$ (and ≥ 2.5 m). Longer bottom layers sometimes used for sliding.
- **Connection** to facing should carry T_{max} (top layers) – check strength of connection and facing.

3.4 Reinforcement layout and spacing

- **Vertical spacing S_v**: 0.3–0.8 m (strips ≈ 0.75 m; geogrid 0.3–0.6 m; wrap-around geotextile 0.3–0.6 m). Smaller spacing → smaller tension per layer and better local stability.
- **Horizontal spacing S_h** (strips): 0.75–1.5 m, matched to panel width; continuous sheets have 100 % coverage.
- First layer at **0.15–0.3 m** above the base; last layer within 0.3 m of the top (below coping).
- Reinforcement laid **horizontally and perpendicular** to the wall face; avoid kinks; no overlap of strips in plan; sheets lapped ≥ 0.3 m.
- Layers may be **lengthened at the top** where surcharge / traffic loads act; length is usually **uniform** with height.

Example 1 – internal

Worked example. $H = 6$ m, $\gamma = 18$ kN/m^3, $\varphi = 32°$ (reinforced and retained fill), geogrid, $S_v = 0.6$ m, no surcharge.

$K_a = (1 - \sin 32°)/(1 + \sin 32°) = 0.307$

Layer at $z = 3$ m: $\sigma_v = 54$ kPa, $\sigma_h = 0.307 \times 54 = 16.6$ kPa → $T = 16.6 \times 0.6 = 9.95$ kN/m

Bottom layer $z = 5.7$ m: $\sigma_v = 102.6$, $\sigma_h = 31.5$ → $T = 18.9$ kN/m → $T_{al} \geq 1.5 \times 18.9 = 28.4$ kN/m

Example 1 – length

Pull-out at z = 3 m: $F^* = (2/3)\tan 32° = 0.417$; $L_e = 1.5 \times 9.95 / (2 \times 0.417 \times 54) = 0.33$ m → adopt 1.0 m (minimum).

$L_a = (6 - 3)\tan 29° = 1.66$ m → $L = 1.66 + 1.0 = 2.7$ m; but $L \geq 0.7H = 4.2$ m (governs) → adopt $L = 4.2$ m.

Example 1 – external

External ($L = 4.2$ m): $W = 18 \times 4.2 \times 6 = 453.6$ kN/m; $P_a = ½ \times 0.307 \times 18 \times 6^2 = 99.6$ kN/m

$FS_{sliding} = 453.6 \times \tan 32° / 99.6 = 283.5/99.6 = 2.85 > 1.5$ OK

$M_R = 453.6 \times 2.1 = 952.6$; $M_O = 99.6 \times 2 = 199.2$ → $FS_o = 4.78 > 2$ OK

$e = 2.1 - (952.6 - 199.2)/453.6 = 0.44$ m $< L/6 = 0.70$ m OK; $q_{max} = 453.6/(4.2 - 0.88) = 136.5$ kPa $\leq q_{all}$ (check)

3.5 Surcharge, live load and seismic effects

- **Uniform surcharge q** (traffic ≈ 10–12 kPa equivalent): increases σ_v and $\sigma_h = K q$ – add to external thrust $K_a q H$ and to tension at every layer.
- **Strip / footing loads** near the wall: spread at 2V : 1H (or Boussinesq) to find extra σ_v and σ_h at each level.
- **Seismic:** pseudo-static (Mononobe–Okabe) – add dynamic thrust ΔP_{AE} on the retained zone and inertia force on the reinforced block; pull-out resistance checked behind the (enlarged) active zone; use longer layers ($L \geq 0.8H$) in high seismic zones.
- **Water:** hydrostatic pressure behind the wall must be avoided by drainage; if unavoidable, use buoyant unit weight and add water thrust.

Example 2 – strength

Example 2 – steel strips. $H = 8$ m, $\varphi = 34°$, $\gamma = 18$, strips 50 × 4 mm galvanised, $S_v = S_h = 0.75$ m, life 75 years.

Corrosion: zinc (85 µm) lasts ≈ 2 yr × 15 + 55/4 ≈ 16 yr; steel loss ≈ 12 µm/yr × 59 yr ≈ 0.71 mm per face → $t_c = 4 - 1.42 = 2.58$ mm

$A_c = 50 \times 2.58 = 129$ mm²; allowable stress $0.55 f_y = 0.55 \times 450 = 247$ MPa → $T_{al} = 31.9$ kN/strip

Example 2 – length

Bottom layer $z = 7.5$ m: $K_a = 0.283$, $K = 1.2 K_a = 0.339$; $\sigma_v = 135$ kPa → $\sigma_h = 45.8$ kPa

$T_{max} = 45.8 \times 0.75 \times 0.75 = 25.8$ kN < 31.9 OK

Pull-out: $F^* = \tan 34° = 0.675$ → $L_e = 1.5 \times 25.8 / (2 \times 0.05 \times 0.675 \times 135) = 4.25$ m

L_a (bilinear) at $z = 7.5$ m = 0.3 m → $L = 4.55$ m; but $L \geq 0.7H = 5.6$ m → adopt $L = 6$ m (rounded).

Example 3

Example 3 – effect of surcharge $q = 12$ kPa on retained fill (Example 1 wall, $L = 4.2$ m):

Extra thrust = $K_a q H = 0.307 \times 12 \times 6 = 22.1$ kN/m acting at $H/2 = 3$ m

$P_{a,total} = 99.6 + 22.1 = 121.7$; $FS_s = 283.5/121.7 = 2.33$ OK

$M_O = 199.2 + 22.1 \times 3 = 265.5$; $FS_o = 952.6/265.5 = 3.59$ OK

Top layer ($z = 0.3$ m): $\sigma_h = 0.307 (18 \times 0.3 + 12) = 5.3$ kPa → $T = 5.3 \times 0.6 = 3.2$ kN/m (higher than without q).

3.6 Facing and connections

- Facing does **not** carry earth pressure – it only retains the face soil between layers and resists local erosion; connection load is ≈ 50–100 % of T_{max} at the top and reduces to ≈ 50–60 % near the base (extensible).
- Precast panels (≈ 1.5 m × 1.5 m) have **joints** with filler and filter strip so that fines do not wash out and differential settlement is accommodated.

- Connections must be corrosion protected (steel) and have creep / strength reduction tested (geosynthetic, block–grid connection).
- Wall **batter** and construction tolerance: ≈ 1° backward batter; alignment ± 20–25 mm.

Short answers

- L_a: length of reinforcement in the active zone (not mobilising resistance). L_e: length in the resistant zone (anchorage length).
- **Why $L \geq 0.7H$?** To provide enough width for the reinforced block to resist sliding, overturning and bearing as a gravity wall.
- **Why different K for steel and geosynthetics?** Inextensible steel restrains lateral strain → higher K at shallow depth.
- **Drainage importance:** a water-filled backfill adds hydrostatic thrust and reduces friction.

4. Construction and Details

1. Prepare and compact the foundation; cast **leveling pad**; set the first facing course plumb.
2. Place reinforced fill in **150–200 mm** lifts; compact ≥ 95 % MDD; light equipment within 1 m of the face.
3. Lay reinforcement flat, pull taut, connect to the facing; fill from the face side moving away.
4. Install drainage and filter geotextile behind the face and at the back of the reinforced zone.
5. Repeat; finish with coping beam, crash barrier, surface drainage and pavement.

- Provide **joints** (compressible filler) in panels to allow settlement; **batter** wall ≈ 1° backward to counter outward movement.
- Control **construction tolerances** – alignment ± 20–25 mm, verticality 1:100.

Common causes of distress

1. **Poor drainage** – water pressure, softened fill (most common cause).
2. **Poor compaction** or use of fine / clayey fill → bulging of the face.
3. **Too short or too weak reinforcement**; corrosion / creep / damage of reinforcement.
4. **Foundation settlement** or bearing failure; scour at the toe (river / sea walls).
5. **Overloading** by unplanned surcharge or construction traffic near the face.

Monitoring: survey targets on facing (outward movement < 1 % H), settlement markers, strain gauges on reinforcement, piezometers and inspection of weep holes.

5. Advantages over Conventional Retaining Walls

Point	Conventional (gravity / cantilever)	Reinforced earth wall
Foundation	Needs strong, deep foundation	Flexible; works on weaker soil
Settlement	Rigid – cracks, tilts	Tolerates large differential settlement
Height	Economical up to ≈ 6–8 m	Economical to 10–30 m
Construction	Formwork, curing – slow	Prefab panels, layer-wise – rapid
Cost	Higher (concrete, steel)	25–50 % cheaper (typical)
Seismic	Rigid, brittle failure	Flexible, energy absorbing
Space / aesthetics	Larger base, one finish	Wide variety of facings

Limitations

- Needs **granular fill** and enough space behind the wall ($L \geq 0.7H$).
- Not suitable where the backfill zone is submerged or contaminated, or for walls with large utilities.
- Corrosion / creep / UV need checking; quality of compaction and drainage is critical.

Formula sheet – Unit III

Item	Expression
Rankine coefficient	$K_a = \tan^2(45^\circ - \varphi/2)$
Active thrust	$P_a = \tfrac{1}{2} K_a \gamma H^2 + K_a q H$
Tension in layer	$T = K_a \sigma_v S_v$ (sheets)
Pull-out length	$L_e \geq FS\, T / (2 b F^* \sigma_v)$
Sliding / overturning	$W \tan\varphi / P_a \geq 1.5$; $M_R / M_O \geq 2$
Eccentricity	$e = L/2 - (M_R - M_O)/W \leq L/6$

Important questions

1. Explain the mechanism of stability of a reinforced earth retaining wall.
2. Describe the design steps of an RE wall; discuss selection of materials.
3. Explain internal and external stability analysis with neat sketches.
4. Explain reinforcement layout and spacing. Compare RE walls with conventional walls.
5. Design an RE wall of 6 m height (given soil data) – numerical problem.